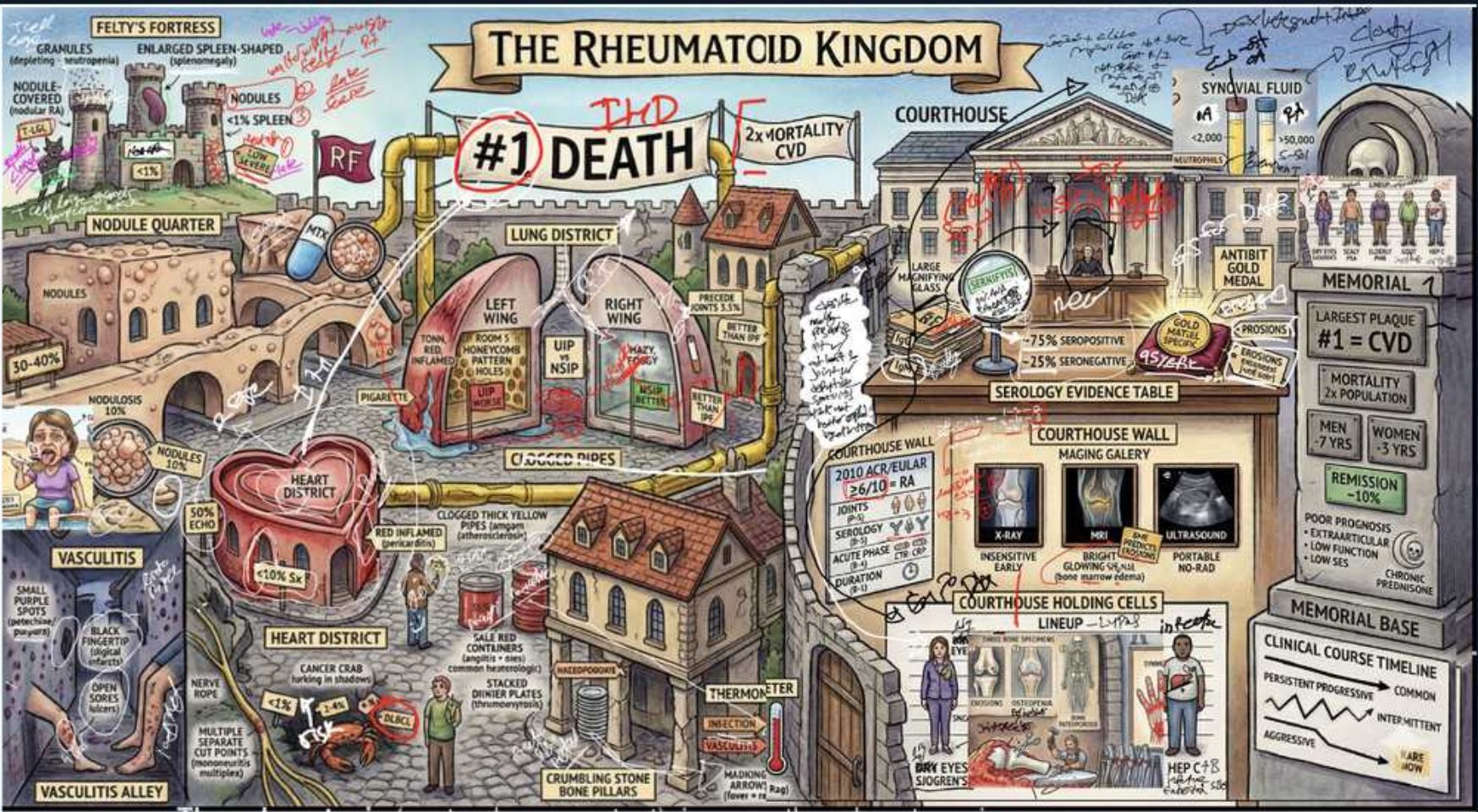


Rheumatoid Arthritis — Extra-articular, Diagnosis & Prognosis



1. Felty syndrome

WHAT YOU SEE

Felty's Fortress: enlarged spleen + granules (neutropenia)

WHAT IT ENCODES

Felty syndrome is the triad of RA + Splenomegaly + Neutropenia (mnemonic 'Felty = SANTA: Splenomegaly, Anemia, Neutropenia, Thrombocytopenia, Arthritis'). It occurs in severe, long-standing, strongly seropositive (high-titer RF and anti-CCP) erosive RA, and the neutropenia (ANC often <1500–2000) markedly raises the risk of recurrent bacterial infections. Treatment is effective DMARD control (methotrexate often raises the neutrophil count); G-CSF for severe infection-associated neutropenia. Large granular lymphocyte (LGL) leukemia is on the differential and shares the picture.

BOARD TRAP

The stem describes seropositive RA with low neutrophils and a palpable spleen, and the wrong answer attributes the cytopenia to drug toxicity or calls it pancytopenia. The discriminator: Felty is defined by NEUTROPENIA specifically (not obligate pancytopenia), and the immediate clinical danger is bacterial infection from the low neutrophil count — not anemia symptoms.

2. Rheumatoid nodules

WHAT YOU SEE

Nodule Quarter / nodule-covered structures on extensor surfaces

WHAT IT ENCODES

Rheumatoid nodules are firm subcutaneous nodules over EXTENSOR/pressure surfaces (olecranon, extensor forearm, Achilles, finger joints); they are the most common cutaneous extra-articular finding. Histology is pathognomonic: a central zone of fibrinoid necrosis surrounded by PALISADING histiocytes/macrophages and an outer rim of lymphocytes. They occur almost exclusively in RF-positive patients and correlate with severe disease and smoking.

BOARD TRAP

A nodule appears or enlarges after starting methotrexate and the wrong answer is 'this is a drug reaction, stop MTX' or 'biopsy to exclude malignancy.' The discriminator: methotrexate can paradoxically trigger ACCELERATED NODULOSIS (small nodules on hands), which is benign and not an indication to stop a working DMARD; and a typical nodule in seropositive RA with palisading granuloma histology does not need a malignancy workup.

3. Vasculitis

WHAT YOU SEE

Vasculitis Alley with skin ulcers and digital lesions

WHAT IT ENCODES

Rheumatoid vasculitis is a small-to-medium vessel vasculitis of severe, long-standing, high-titer seropositive RA. Manifestations: nailfold/periungual infarcts, palpable purpura, deep cutaneous ulcers, digital ischemia/gangrene, and mononeuritis multiplex (vasculitis of the vasa nervorum causing patchy motor/sensory deficits like wrist or foot drop). It signals aggressive disease and is treated with high-dose steroids plus an immunosuppressant (e.g., rituximab or cyclophosphamide).

BOARD TRAP

A long-standing seropositive RA patient develops a sudden foot drop or wrist drop, and the wrong answer is 'compressive mononeuropathy' or 'worsening arthritis.' The discriminator: asymmetric, stepwise involvement of named nerves = mononeuritis multiplex from rheumatoid VASCULITIS, a systemic emergency requiring immunosuppression, not a mechanical/local explanation.

4. ILD (lung district)

WHAT YOU SEE

Lung District, UIP > NSIP patterns marked

WHAT IT ENCODES

RA-associated interstitial lung disease presents with progressive dyspnea, dry cough, and bibasilar 'Velcro' crackles. The most common HRCT/histologic pattern is UIP (usual interstitial pneumonia: basal, subpleural reticulation, honeycombing, traction bronchiectasis), which — like idiopathic pulmonary fibrosis — carries the WORST prognosis; NSIP (more ground-glass) is the second pattern and responds better to immunosuppression. Male sex, smoking, older age, and high anti-CCP titers increase risk.

BOARD TRAP

Attributing a fibrotic basal lung picture in an RA patient solely to methotrexate. The discriminator: methotrexate causes an ACUTE hypersensitivity pneumonitis, whereas chronic basal fibrosis with honeycombing is RA-ILD (UIP pattern) itself; mislabeling it as drug toxicity (and missing the UIP = worse prognosis point) is the trap. Always also exclude infection on immunosuppression.

5. Caplan syndrome

WHAT YOU SEE

Lung wings with nodules in a pneumoconiosis setting

WHAT IT ENCODES

Caplan syndrome (rheumatoid pneumoconiosis) is the combination of RA + inhaled mineral dust exposure (COAL, silica, asbestos) producing multiple well-defined upper/mid-zone pulmonary NODULES that can cavitate. Classic stem: a coal MINER or quarry/foundry worker with seropositive RA and rounded lung nodules on imaging.

BOARD TRAP

Multiple cavitating lung nodules in a coal miner with RA, and the wrong answer is 'metastatic malignancy' or 'reactivation TB.' The discriminator: the occupational dust exposure plus seropositive RA points to Caplan nodules (rheumatoid pneumoconiosis); the history of dust inhalation is the giveaway, though TB/malignancy must still be excluded clinically.

6. Pericarditis / heart district

WHAT YOU SEE

Heart District with pericardial fluid containers

WHAT IT ENCODES

Pericarditis is the most common CARDIAC manifestation of RA and is frequently asymptomatic, found as an exudative pericardial effusion on echo. RA also accelerates coronary atherosclerosis, can cause myocarditis, valvular nodules, and conduction disease. Symptomatic pericarditis is treated with NSAIDs/steroids and DMARD optimization; rarely it progresses to constrictive pericarditis or tamponade.

BOARD TRAP

Equating 'most common cardiac manifestation' (pericarditis) with 'leading cause of cardiac DEATH' in RA. The discriminator: pericarditis is the most common cardiac finding, but the number-one cause of mortality in RA is accelerated coronary/ischemic CARDIOVASCULAR disease, not pericarditis itself.

7. Cardiovascular death = #1 cause

WHAT YOU SEE

#1 DEATH banner, 2x mortality CVD

WHAT IT ENCODES

Cardiovascular disease is the LEADING cause of death in RA, with roughly a 2-fold increased risk driven by chronic systemic inflammation accelerating atherosclerosis (an independent risk factor beyond traditional ones). This is why tight inflammatory control with DMARDs (methotrexate is cardioprotective) and aggressive modification of traditional risk factors (lipids, blood pressure, smoking cessation) are central to RA care.

BOARD TRAP

A board question asks for the leading cause of death/excess mortality in RA and the distractors are 'infection,' 'the joint disease,' 'lymphoma,' or 'amyloidosis.' The discriminator: accelerated CARDIOVASCULAR/ischemic heart disease (~2x risk) is the number-one killer; infection and malignancy raise mortality but are not the top cause.

8. RF (rheumatoid factor)

WHAT YOU SEE

A red 'RF' label tagged on the Serology Evidence Table — RF is the sensitive-but-not-specific antibody, the wide net that catches many diseases, not just RA.

WHAT IT ENCODES

Rheumatoid factor is an autoantibody (usually IgM) directed against the Fc portion of IgG. It is fairly SENSITIVE (~70–80%) but NOT specific for RA — it is also positive in Sjögren syndrome (very high), chronic infections (hepatitis C, subacute bacterial endocarditis, TB), other connective tissue diseases, cryoglobulinemia, and healthy elderly people. Higher titers correlate with more severe, erosive, extra-articular disease.

BOARD TRAP

A positive RF is offered as 'diagnostic of RA' or used to diagnose RA in a patient with hepatitis C, Sjögren, or endocarditis. The discriminator: RF lacks specificity — anti-CCP (~95% specific) is the confirmatory antibody. A positive RF in the context of HCV/SBE/Sjögren may reflect that disease, not RA.

9. Anti-CCP (ACPA)

WHAT YOU SEE

The Serology Evidence Table singling out anti-CCP as the prosecution's strongest evidence — it is the ~95%-specific 'smoking gun' antibody that nails the RA diagnosis where RF cannot.

WHAT IT ENCODES

Anti-cyclic citrullinated peptide antibody (anti-CCP / ACPA) is the most SPECIFIC serology for RA (~95% specific, sensitivity similar to RF ~70%). It can precede clinical arthritis by years, predicts an erosive/aggressive course, and is a poor-prognosis marker. In the 2010 ACR/EULAR classification criteria, high-titer anti-CCP (or RF) contributes the most points toward classifying RA.

BOARD TRAP

Choosing RF over anti-CCP when the vignette emphasizes a confounding RF-positive condition (HCV, Sjögren) and asks for the MOST SPECIFIC test, or claiming anti-CCP is the most sensitive. The discriminator: anti-CCP = high SPECIFICITY (~95%) and prognostic significance; RF = higher sensitivity but poor specificity. Confusing sensitivity with specificity is the trap.

10. AA amyloidosis

WHAT YOU SEE

A graveyard MEMORIAL with headstones/skulls fed by chronic inflammation — chronic uncontrolled RA seeds serum amyloid A deposits that bury the kidneys (AA amyloidosis).

WHAT IT ENCODES

Chronic uncontrolled inflammation in RA can cause SECONDARY (AA) amyloidosis: serum amyloid A (an acute-phase reactant) deposits in organs, classically the KIDNEY, causing nephrotic-range proteinuria and progressive renal failure. Diagnosis is by biopsy (fat pad/rectal/renal) with Congo red stain showing apple-green birefringence under polarized light. Best prevention/treatment is suppressing the underlying inflammation.

BOARD TRAP

A long-standing poorly controlled RA patient with new nephrotic-range proteinuria, and the wrong answer is 'AL (primary) amyloidosis' or 'membranous nephropathy.' The discriminator: chronic inflammatory disease (RA, IBD, chronic infection) causes AA (serum amyloid A) amyloidosis, NOT AL (light-chain) amyloid; AL is associated with plasma cell dyscrasias/myeloma instead.

11. X-ray / imaging gallery

WHAT YOU SEE

Courthouse Wall Imaging Gallery: X-ray, MRI, ultrasound

WHAT IT ENCODES

Plain radiographs show the classic RA triad: symmetric joint-space narrowing, MARGINAL (juxta-articular) bony erosions, and PERIARTICULAR osteopenia — typically in the MCPs, PIPs, wrists, and MTPs. Early in disease X-rays may be normal; MRI and musculoskeletal ULTRASOUND (with power Doppler) detect synovitis and erosions far earlier and guide early DMARD initiation within the 'window of opportunity.'

BOARD TRAP

Excluding early RA because hand X-rays are normal. The discriminator: radiographs are insensitive early — a normal X-ray does NOT rule out active erosive synovitis; MRI/ultrasound (power Doppler) reveal early disease, and treatment shouldn't wait for radiographic erosions to appear. Also distinguish RA's marginal erosions/periarticular osteopenia from OA's osteophytes/subchondral sclerosis.

12. Scleritis / eye lookups

WHAT YOU SEE

A police-lineup of RED-EYED suspects in the courthouse holding cells — the red eye that is deeply PAINFUL is the dangerous suspect (scleritis), versus the painless mild-red episcleritis.

WHAT IT ENCODES

Ocular RA: the MOST COMMON is keratoconjunctivitis sicca (dry eyes from secondary Sjögren). The vascular spectrum runs from benign episcleritis to sight-threatening scleritis. SCLERITIS is deep, severely painful (boring pain, painful eye movement), with a violaceous/bluish hue and possible scleral thinning/necrosis (scleromalacia perforans) — an ophthalmologic emergency needing systemic immunosuppression. Episcleritis is superficial, mildly red, NON-painful, and self-limited.

BOARD TRAP

A red eye in an RA patient where the wrong answer treats painful scleritis as benign episcleritis (or vice versa). The discriminator: PAIN and vision threat = scleritis (needs systemic steroids/immunosuppression and urgent ophthalmology); painless mild redness = episcleritis (benign, topical care). Phenylephrine blanches episcleral vessels but not the deep scleral vessels — a clue to differentiate.

13. Poor prognosis markers

WHAT YOU SEE

Memorial Base: high RF/anti-CCP titer, early erosions, extra-articular disease

WHAT IT ENCODES

Poor-prognosis features in RA: high-titer RF AND/OR anti-CCP positivity, early/numerous bony erosions, high disease activity with many involved joints, presence of extra-articular manifestations (nodules, ILD, vasculitis), persistently elevated ESR/CRP, functional limitation, and the HLA-DRB1 shared epitope. These features justify earlier, more aggressive DMARD/biologic therapy and tighter treat-to-target follow-up.

BOARD TRAP

A vignette pairs seropositivity, early erosions, and nodules and asks for prognosis/management, with the wrong answer being 'mild disease, start NSAIDs and reassess.' The discriminator: clustering of poor-prognosis markers (high-titer antibodies + erosions + extra-articular disease) mandates EARLY aggressive DMARD therapy, not symptomatic NSAID-only management.

14. Remission ~10% / clinical course

WHAT YOU SEE

Clinical Course Timeline with low remission percentage

WHAT IT ENCODES

Untreated RA is typically a chronic, PROGRESSIVE disease — spontaneous sustained remission is uncommon. Early DMARD therapy within the 'window of opportunity' (first months) dramatically alters the trajectory, preventing irreversible erosions and disability. The course is usually a persistent or relapsing-remitting polyarthritis rather than a self-limited illness, which is why prompt diagnosis and treat-to-target management matter.

BOARD TRAP

Assuming RA will 'burn out' or remit on its own, leading to a wait-and-watch approach. The discriminator: RA is progressive and joint damage is irreversible — delaying DMARDs to see if it resolves squanders the window of opportunity and worsens long-term outcomes. Early aggressive treatment, not observation, is correct.